

## SAFETY DATA SHEET

Creation Date 22-September-2009

Revision Date 26-March-2024

Revision Number 3

### 1. Identification

**Product Name** Silver wire

**Cat No. :** 41455

**CAS-No** 7440-22-4  
**Synonyms** Argentum.

**Recommended Use** Laboratory chemicals.  
**Uses advised against** Food, drug, pesticide or biocidal product use.

#### Details of the supplier of the safety data sheet

##### Company

##### **Importer/Distributor**

Fisher Scientific  
112 Colonnade Road,  
Ottawa, ON K2E 7L6,  
Canada  
Tel: 1-800-234-7437

##### **Emergency Telephone Number**

For information **US** call: 001-800-227-6701 / **Europe** call: +32 14 57 52 11

Emergency Number **US**:001-201-796-7100 / **Europe**: +32 14 57 52 99

**CHEMTREC** Tel. No. **US**:001-800-424-9300 / **Europe**:001-703-527-3887

### 2. Hazard(s) identification

#### Classification

**WHMIS 2015 Classification** Not classified under the Hazardous Products Regulations (SOR/2015-17)

Based on available data, the classification criteria are not met

#### Label Elements

None required

### 3. Composition/Information on Ingredients

| Component | CAS-No    | Weight % |
|-----------|-----------|----------|
| Silver    | 7440-22-4 | 100      |

#### 4. First-aid measures

|                                                               |                                                                                                                              |
|---------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|
| <b>Eye Contact</b>                                            | Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get medical attention.              |
| <b>Skin Contact</b>                                           | Wash off immediately with soap and plenty of water while removing all contaminated clothes and shoes. Get medical attention. |
| <b>Inhalation</b>                                             | Remove to fresh air. Get medical attention. If not breathing, give artificial respiration.                                   |
| <b>Ingestion</b>                                              | Do NOT induce vomiting. Get medical attention.                                                                               |
| <b>Most important symptoms/effects<br/>Notes to Physician</b> | No information available.<br>Treat symptomatically                                                                           |

#### 5. Fire-fighting measures

|                                         |                                                                                                                                                                                      |
|-----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Suitable Extinguishing Media</b>     | Water spray. Carbon dioxide (CO <sub>2</sub> ). Dry chemical. Use extinguishing measures that are appropriate to local circumstances and the surrounding environment. Chemical foam. |
| <b>Unsuitable Extinguishing Media</b>   | No information available                                                                                                                                                             |
| <b>Flash Point</b>                      | No information available                                                                                                                                                             |
| <b>Method -</b>                         | No information available                                                                                                                                                             |
| <b>Autoignition Temperature</b>         | No information available                                                                                                                                                             |
| <b>Explosion Limits</b>                 |                                                                                                                                                                                      |
| <b>Upper</b>                            | No data available                                                                                                                                                                    |
| <b>Lower</b>                            | No data available                                                                                                                                                                    |
| <b>Sensitivity to Mechanical Impact</b> | No information available                                                                                                                                                             |
| <b>Sensitivity to Static Discharge</b>  | No information available                                                                                                                                                             |

#### Specific Hazards Arising from the Chemical

Non-combustible.

#### Hazardous Combustion Products

None known.

#### Protective Equipment and Precautions for Firefighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

#### NFPA

**Health**  
1

**Flammability**  
0

**Instability**  
0

**Physical hazards**  
N/A

#### 6. Accidental release measures

|                                  |                                                                             |
|----------------------------------|-----------------------------------------------------------------------------|
| <b>Personal Precautions</b>      | Ensure adequate ventilation. Use personal protective equipment as required. |
| <b>Environmental Precautions</b> | Do not flush into surface water or sanitary sewer system.                   |

**Methods for Containment and Clean Up** Sweep up and shovel into suitable containers for disposal.

#### 7. Handling and storage

|                 |                                                                                                                                                               |
|-----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Handling</b> | Ensure adequate ventilation. Wear personal protective equipment/face protection. Avoid contact with skin and eyes. Do not breathe dust. Avoid dust formation. |
| <b>Storage.</b> | Keep in a dry, cool and well-ventilated place. Keep container tightly closed. Incompatible                                                                    |

Materials. Acids. Strong acids. Strong bases. Ammonia. oxygen.

## 8. Exposure controls / personal protection

### Exposure Guidelines

| Component | Alberta                    | British Columbia                                            | Ontario TWAEV              | Quebec                     | ACGIH TLV                  | OSHA PEL                                                             | NIOSH                                                                                   |
|-----------|----------------------------|-------------------------------------------------------------|----------------------------|----------------------------|----------------------------|----------------------------------------------------------------------|-----------------------------------------------------------------------------------------|
| Silver    | TWA: 0.1 mg/m <sup>3</sup> | TWA: 0.01 mg/m <sup>3</sup><br>STEL: 0.03 mg/m <sup>3</sup> | TWA: 0.1 mg/m <sup>3</sup> | TWA: 0.1 mg/m <sup>3</sup> | TWA: 0.1 mg/m <sup>3</sup> | (Vacated) TWA: 0.01 mg/m <sup>3</sup><br>TWA: 0.01 mg/m <sup>3</sup> | IDLH: 10 mg/m <sup>3</sup><br>TWA: 0.01 mg/m <sup>3</sup><br>TWA: 0.9 µg/m <sup>3</sup> |

#### Legend

ACGIH - American Conference of Governmental Industrial Hygienists

OSHA - Occupational Safety and Health Administration

NIOSH: NIOSH - National Institute for Occupational Safety and Health

### Engineering Measures

None under normal use conditions.

### Personal protective equipment

#### Eye Protection

Wear safety glasses with side shields (or goggles)

#### Hand Protection

Protective gloves

| Glove material | Breakthrough time                 | Glove thickness | Glove comments         |
|----------------|-----------------------------------|-----------------|------------------------|
| Natural rubber | See manufacturers recommendations | -               | Splash protection only |
| Nitrile rubber |                                   |                 |                        |
| Neoprene       |                                   |                 |                        |
| PVC            |                                   |                 |                        |

Inspect gloves before use. observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information) gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion. gloves with care avoiding skin contamination.

#### Respiratory Protection

No protective equipment is needed under normal use conditions.

**Recommended Filter type:** Particle filter

### Environmental exposure controls

Prevent product from entering drains. Do not allow material to contaminate ground water system. Local authorities should be advised if significant spillages cannot be contained.

### Hygiene Measures

Handle in accordance with good industrial hygiene and safety practice. Keep away from food, drink and animal feeding stuffs. Do not eat, drink or smoke when using this product. Remove and wash contaminated clothing and gloves, including the inside, before re-use. Wash hands before breaks and after work.

## 9. Physical and chemical properties

#### Physical State

Solid

#### Appearance

Silver

#### Odor

Odorless

#### Odor Threshold

No information available

#### pH

No information available

#### Melting Point/Range

961 °C / 1761.8 °F

#### Boiling Point/Range

2212 °C / 4013.6 °F

|                                        |                          |
|----------------------------------------|--------------------------|
| Flash Point                            | No information available |
| Evaporation Rate                       | Not applicable           |
| Flammability (solid,gas)               | No information available |
| Flammability or explosive limits       |                          |
| Upper                                  | No data available        |
| Lower                                  | No data available        |
| Vapor Pressure                         | No information available |
| Vapor Density                          | Not applicable           |
| Specific Gravity                       | 10.5                     |
| Solubility                             | insoluble                |
| Partition coefficient; n-octanol/water | No data available        |
| Autoignition Temperature               | No information available |
| Decomposition Temperature              | No information available |
| Viscosity                              | Not applicable           |
| Molecular Formula                      | Ag                       |
| Molecular Weight                       | 107.87                   |

## 10. Stability and reactivity

|                                  |                                                    |
|----------------------------------|----------------------------------------------------|
| Reactive Hazard                  | None known, based on information available         |
| Stability                        | Air sensitive.                                     |
| Conditions to Avoid              | Exposure to air. Incompatible products.            |
| Incompatible Materials           | Acids, Strong acids, Strong bases, Ammonia, oxygen |
| Hazardous Decomposition Products | None under normal use conditions                   |
| Hazardous Polymerization         | Hazardous polymerization does not occur.           |
| Hazardous Reactions              | None under normal processing.                      |

## 11. Toxicological information

### Acute Toxicity

**Product Information** No acute toxicity information is available for this product

#### Component Information

| Component | LD50 Oral            | LD50 Dermal               | LC50 Inhalation              |
|-----------|----------------------|---------------------------|------------------------------|
| Silver    | > 2000 mg/kg ( Rat ) | LD50 > 2000 mg/kg ( rat ) | LC50 > 5.16 mg/L ( Rat ) 4 h |

**Toxicologically Synergistic Products** No information available

### Delayed and immediate effects as well as chronic effects from short and long-term exposure

|                 |                                                                                          |
|-----------------|------------------------------------------------------------------------------------------|
| Irritation      | No information available                                                                 |
| Sensitization   | No information available                                                                 |
| Carcinogenicity | The table below indicates whether each agency has listed any ingredient as a carcinogen. |

| Component | CAS-No    | IARC       | NTP        | ACGIH      | OSHA       | Mexico     |
|-----------|-----------|------------|------------|------------|------------|------------|
| Silver    | 7440-22-4 | Not listed | Not listed | Not listed | Not listed | Not listed |

**Mutagenic Effects** No information available

**Reproductive Effects** No information available.

**Developmental Effects** No information available.

**Teratogenicity** No information available.

STOT - single exposure None known  
STOT - repeated exposure None known

Aspiration hazard No information available

Symptoms / effects, both acute and delayed No information available

Endocrine Disruptor Information No information available

Other Adverse Effects The toxicological properties have not been fully investigated.

## 12. Ecological information

### Ecotoxicity

Contains a substance which is: Very toxic to aquatic organisms. The product contains following substances which are hazardous for the environment.

| Component | Freshwater Algae | Freshwater Fish                                                                                                                                                                       | Microtox   | Water Flea                                       |
|-----------|------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|--------------------------------------------------|
| Silver    | Not listed       | LC50: = 0.064 mg/L, 96h static (Lepomis macrochirus)<br>LC50: = 0.0062 mg/L, 96h flow-through (Oncorhynchus mykiss)<br>LC50: 0.00155 - 0.00293 mg/L, 96h static (Pimephales promelas) | Not listed | EC50: = 0.00024 mg/L, 48h Static (Daphnia magna) |

Persistence and Degradability Insoluble in water

Bioaccumulation/ Accumulation No information available.

Mobility Is not likely mobile in the environment due its low water solubility.

## 13. Disposal considerations

Waste Disposal Methods Chemical waste generators must determine whether a discarded chemical is classified as a hazardous waste. Chemical waste generators must also consult local, regional, and national hazardous waste regulations to ensure complete and accurate classification.

## 14. Transport information

DOT Not regulated  
TDG Not regulated  
IATA Not regulated  
IMDG/IMO Not regulated

## 15. Regulatory information

### International Inventories

| Component | CAS-No    | DSL | NDSL | TSCA | TSCA Inventory notification - Active-Inactive | EINECS    | ELINCS | NLP |
|-----------|-----------|-----|------|------|-----------------------------------------------|-----------|--------|-----|
| Silver    | 7440-22-4 | X   | -    | X    | ACTIVE                                        | 231-131-3 | -      | -   |

  

| Component | CAS-No    | IECSC | KECL     | ENCS | ISHL | TCSI | AICS | NZIoC | PICCS |
|-----------|-----------|-------|----------|------|------|------|------|-------|-------|
| Silver    | 7440-22-4 | X     | KE-31261 | X    | -    | X    | X    | X     | X     |

### Legend:

X - Listed '-' - Not Listed

KECL - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

DSL/NDL - Canadian Domestic Substances List/Non-Domestic Substances List

TSCA - United States Toxic Substances Control Act Section 8(b) Inventory

EINECS/ELINCS - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

IECSC - Chinese Inventory of Existing Chemical Substances

KECL - Korean Existing and Evaluated Chemical Substances

ENCS - Japanese Existing and New Chemical Substances

AICS - Australian Inventory of Chemical Substances

PICCS - Philippines Inventory of Chemicals and Chemical Substances

## Canada

SDS in compliance with provisions of information as set out in Canadian Standard - Part 4, Schedule 1 and 2 of the Hazardous Products Regulations (HPR) and meets the requirements of the HPR (Paragraph 13(1)(a) of the Hazardous Products Act (HPA)).

| Component | Canada - National Pollutant Release Inventory (NPRI) | Canadian Environmental Protection Agency (CEPA)<br>- List of Toxic Substances | Canada's Chemicals Management Plan (CEPA) |
|-----------|------------------------------------------------------|-------------------------------------------------------------------------------|-------------------------------------------|
| Silver    | Part 1, Group A Substance                            |                                                                               |                                           |

## Other International Regulations

### Authorisation/Restrictions according to EU REACH

| Component | REACH (1907/2006) - Annex XIV - Substances Subject to Authorization | REACH (1907/2006) - Annex XVII - Restrictions on Certain Dangerous Substances | REACH Regulation (EC 1907/2006) article 59 - Candidate List of Substances of Very High Concern (SVHC) |
|-----------|---------------------------------------------------------------------|-------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| Silver    | -                                                                   | Use restricted. See item 75.<br>(see link for restriction details)            | -                                                                                                     |

### REACH links

<https://echa.europa.eu/substances-restricted-under-reach>

### Safety, health and environmental regulations/legislation specific for the substance or mixture

| Component | CAS-No    | OECD HPV | Persistent Organic Pollutant | Ozone Depletion Potential | Restriction of Hazardous Substances (RoHS) |
|-----------|-----------|----------|------------------------------|---------------------------|--------------------------------------------|
| Silver    | 7440-22-4 | Listed   | Not applicable               | Not applicable            | Not applicable                             |

| Component | CAS-No    | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements | Rotterdam Convention (PIC) | Basel Convention (Hazardous Waste) |
|-----------|-----------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|----------------------------|------------------------------------|
| Silver    | 7440-22-4 | Not applicable                                                                            | Not applicable                                                                           | Not applicable             | Not applicable                     |

## 16. Other information

### Prepared By

Product Safety Department  
Email: chem.techinfo@thermofisher.com  
www.thermofisher.com

### Creation Date

22-September-2009

### Revision Date

26-March-2024

### Print Date

26-March-2024

### Revision Summary

New emergency telephone response service provider.

### Disclaimer

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage,

transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of SDS**